Two HRS Core cognitive classification methods informed by HRS/HCAP

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Abstract

- **Importance:** population cognitive classification in HRS Core; dementia and MCI surveillance; limited assessment battery
- **Objective:** describe two HRS Core methods calibrated to HRS/HCAP (Core algorithm, PMM); compare with the HRS/HCAP Algorithm, consensus, Langa-Weir, and Hudomiet
- **Design:** 2016 HRS Core classification in HRS/HCAP participants; consensus-panel secondary comparator
    - Study sample: HRS/HCAP 2016 participants, N = 3,496
    - [Provenance: R/MS_MAIN/MS_001_analysis_cohort.R]
- **Methods:** Core algorithm; PMM
    - Primary comparator: HRS/HCAP Algorithm
    - Secondary comparators: (1) HCAP consensus; (2) Langa-Weir; (3) Hudomiet classification
    - Statistical analysis: dementia versus non-dementia kappa; 3-class quadratic weighted kappa (Normal, MCI, Dementia); all statistics other than frequencies use complex sampling weights
    - [Provenance: R/MS_MAIN/MS_010_results_objects.R]
- **Results:**
    - Dementia/non-dementia kappa
        - Core algorithm versus HRS/HCAP Algorithm: 0.337
        - PMM versus HRS/HCAP Algorithm: 0.429
    - 3-class kappa
        - Core algorithm versus HRS/HCAP Algorithm: 0.501
        - PMM versus HRS/HCAP Algorithm: 0.509
    - [Provenance: R_objects/A7_100_hcap_tables.rds; R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]
- **Conclusions and relevance:** Two core classification approaches; agreement evidence rather than clinical validation; uncertainty in distinguishing MCI from normal cognition and dementia; limited precision of comparisons with consensus diagnoses because the consensus-panel subsample is small
    - [Provenance: analysis design; inference]

Key terms

- **HRS/HCAP Algorithm**: c.f. Manly, Jones et al. (2022)
- **Core algorithm**: HCAP algorithm adapted to the HRS Core data

- **PMM**: Profile mixture model using measures collected in the HRS Core, calibrated to the HRS/HCAP Algorithm and scored in the HRS/HCAP sample

Introduction

- Population cognitive classification from brief HRS Core assessments
    - Prevalence estimation; etiologic research; trial sampling; policy surveillance
    - Core assessment constraints; incomplete cognitive testing; proxy/informant information
- HRS/HCAP Algorithm as primary comparator
    - HCAP neuropsychological assessment; actuarial Normal/MCI/Dementia classification
    - HRS/HCAP Algorithm; primary HCAP-based comparator
    - Manly et al. (2022); Farron et al. (2025)
    - [Provenance: R/A7_100-comparison_of_diagnoses.R; existing manuscript materials]
- Study aim
    - Core algorithm and PMM; HRS/HCAP Algorithm primary comparator
    - Secondary comparators: HCAP consensus; Langa-Weir; Hudomiet
    - Classification margins; 3-class agreement; dementia/non-dementia agreement

Methods

Design and analytic samples

- Study sample: HRS/HCAP 2016 participants, N = 3,496
    - All HCAP participants; no Core cognitive-test requirement
    - Participants without Core cognitive testing classified from informant data (Jorm IQCODE): Core algorithm Jorm fallback; PMM Jorm model
    - Agreement-statistic samples: Core algorithm N = 3,496; PMM N = 3,496
    - [Provenance: R/MS_MAIN/MS_001_analysis_cohort.R; R_objects/A7_005_hrs16_merged.rds; R/A7_100-comparison_of_diagnoses.R; R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]
- Consensus panel subset
    - Validation-subsample design; inverse-probability weighting to account for selection into the consensus panel
    - [Provenance: R/A7_100-comparison_of_diagnoses.R; R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]

HRS Core classification approaches

- Two Core classification approaches
    - Core algorithm
        - Survey-weighted CFA; demographically normed cognitive score; cognitive/self-report/informant rules

        - [Provenance: R/A7_050-mplus_model.R; R/A7_055-norm_factor_score.R; R/A7_075-implement_algorithm.R]
    - PMM
        - HRS/HCAP-calibrated probabilistic Normal/MCI/Dementia classification; combined cognition and Jorm models (i.e., two PMM models, one trained using cognition data, the other trained using informant data (Jorm IQCODE)).
        - *Limitation*: The HRS/HCAP IQCODE look back window is the past 10 years, the Core IQCODE has a look back window of 2 years.
        - [Provenance: R/PMM_103_calibration_model.R; R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]

- Single-factor Core cognition model
    - Not enough items to separately score 5 domains matching Manly et al. (2022)
    - Initial 8-indicator, complex-survey model fit evaluated with WLSMV
    - Factor scores estimated with MLR, a probit link, and EAP scoring after fixing item parameters to the MLR model estimates
    - Factor-score normalization; age, sex, race/ethnicity, education covariates
    - [Provenance: R/A7_050-mplus_model.R; mplus_output/A7/model_1a_fixed/model_1a_fixed.inp; R/A7_055-norm_factor_score.R]
- Rule-based classification
    - Cognitive thresholds; self-reported concern; JORM for missing cognitive assessments
    - Normal, MCI, Dementia output
    - More detail in **Appendix 1**
    - [Provenance: R/A7_075-implement_algorithm.R]

- HRS/HCAP calibration
    - Known-class HRS/HCAP Algorithm classification; cognitive, functional, self-report, informant indicators
    - No latent variables for overall cognition, individual tests used
    - Calibration parameters fixed for scoring; scored in the same HRS/HCAP sample used for calibration (in-sample classification); not yet applied to the HRS Core sample outside HCAP
    - [Provenance: R/PMM_100_norming_npb.R; R/PMM_103_calibration_model.R]
- Class proportions (priors)

    - Class-specific parameters estimated with HCAP16WGTR; MLR estimation, Bayes-rule posterior classification (plug-in rule; McLachlan, 1992)
    - Class proportions fixed at the HCAP16WGTR-weighted HRS/HCAP Algorithm distribution, computed outside Mplus; Mplus KNOWNCLASS estimates class proportions from unweighted counts
        - Cognition model: Normal 69.4%, MCI 21.4%, Dementia 9.2%
        - Jorm model: Normal 31.7%, MCI 31.6%, Dementia 36.7%
    - [Provenance: R/PMM_101_mplus_function.R; R/PMM_103_calibration_model.R; R/PMM_104_weight_check.R; mplus_output/pmm_103/pmm_hcap_103b.out; mplus_output/pmm_103_jorm/pmm_hcap_103b_jorm.out]
- Combined classification
    - Cognition-model probabilities; JORM-model probabilities when available
    - Probabilistic Normal/MCI/Dementia membership; replicated probability-weighted comparisons
    - More detail in **Appendix 2**
    - [Provenance: R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]

Comparison classifications and estimands

- Primary comparator: HRS/HCAP Algorithm
    - HCAP population weights: HCAP16WGTR
    - [Provenance: R/A7_100-comparison_of_diagnoses.R]
- Secondary comparators
    - HCAP consensus diagnosis: normalized HCAP16WGTR x samplingP^-1 weights
    - Langa-Weir; Hudomiet classification
    - [Provenance: R/A0_017-recode_classification.R; R/A7_100-comparison_of_diagnoses.R]
- Agreement estimands
    - 3-class quadratic weighted kappa
    - Dementia versus non-dementia kappa
    - Marginal Normal/MCI/Dementia distributions
    - [Provenance: R/MS_MAIN/MS_010_results_objects.R]

Results

Analytic samples and classification margins

- Table 1: HRS/HCAP study sample characteristics
    - Dynamic counts and descriptive statistics

    - [Provenance: R/MS_MAIN/MS_Tab_Fig_Apndx-030-Table-1.qmd]
- Table 2: weighted classification margins
    - HRS/HCAP Algorithm; Core algorithm; PMM; Langa-Weir; Hudomiet
    - HCAP16WGTR-weighted margins
        - HRS/HCAP Algorithm: Normal 69.4%, MCI 21.4%, Dementia 9.2%
        - Core algorithm: Normal 69.6%, MCI 21.2%, Dementia 9.1%
        - PMM: Normal 76.7%, MCI 13.6%, Dementia 9.7%
    - [Provenance: R/MS_MAIN/MS_010_results_objects.R]
- Figure 1: Core algorithm
    - Classification rules and participant counts
    - [Provenance: R/MS_MAIN/MS_Tab_Fig_Apndx-020-Figure-1.qmd]
- Figure 2: PMM class-probability distributions
    - Probability separation; uncertainty near class boundaries
    - [Provenance: R/PMM_112_Margins.R]

Agreement with HRS/HCAP Algorithm

- Core algorithm versus HRS/HCAP Algorithm: 3-class kappa 0.501; binary kappa 0.337
    - [Provenance: R_objects/A7_100_hcap_tables.rds]
- PMM versus HRS/HCAP Algorithm: 3-class kappa 0.509; binary kappa 0.429
    - [Provenance: R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]
- Table 3: pairwise agreement, comparators, weight definitions
    - [Provenance: R/MS_MAIN/MS_010_results_objects.R]

Agreement with consensus and established HRS classifications

- Core algorithm, PMM, Langa-Weir, Hudomiet
    - Separate HCAP population and consensus-subsample estimands
    - Weight-dependent interpretation
    - [Provenance: R/MS_MAIN/MS_010_results_objects.R]
- Table 4: A7 pairwise three-class agreement matrix
    - [Provenance: R_objects/A7_100_hcap_tables.rds]

Discussion

- Two HRS Core classification approaches
    - Rule-based Core algorithm; probabilistic PMM classification
    - Same HRS/HCAP learning target; distinct measurement and classification assumptions

    - [Provenance:analysis methods]
- Primary comparison
    - HRS/HCAP Algorithm; agreement and marginal-distribution evidence
    - [Provenance: Tables 2-4]
- Secondary comparisons
    - Consensus: small validation subset; inverse-probability weights
    - Langa-Weir and Hudomiet: established HRS comparators; different development targets
    - [Provenance: R/A7_100-comparison_of_diagnoses.R]
- Interpretation limits
    - Classification agreement; no clinical diagnostic validation outside HCAP consensus subset
    - MCI boundary; model-dependent and threshold-dependent assignments
    - PMM agreement with the HRS/HCAP Algorithm is in-sample (apparent error rate; McLachlan, 1992); expected to be optimistic relative to application in new samples
    - [Provenance: analysis design; inference; R/PMM_103_calibration_model.R]

Two HRS Core cognitive classification methods informed by HRS/HCAP: Figures, tables, and appendices

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Table 1. Participant Characteristics

Characteristic	HRS/HCAP 2016 participants (N = 3,496)
Sex (N (%))	
Female	2,095 (59.9%)
Male	1,401 (40.1%)
Age in years (Mean (SD))	76.0 (7.6)
Race (N (%))	
White	2,755 (78.8%)
Black	563 (16.1%)
All other groups	177 (5.1%)
Not collected	1 (0.0%)
Hispanic origin (N (%))	
Not Hispanic	3,111 (89.0%)
Hispanic	383 (11.0%)
Not obtained	2 (0.1%)
Years of education (Mean (SD))	12.7 (3.2)
HRS/HCAP Classification (N (%))	
Normal	2,313 (66.2%)
Mild cognitive impairment	808 (23.1%)
Dementia	375 (10.7%)

Table 1. Participant Characteristics

Characteristic	HRS/HCAP 2016 participants (N = 3,496)
Langa-Weir Classification (N (%))	
Normal	2,432 (69.6%)
Cognitively impaired, not demented	731 (20.9%)
Dementia	333 (9.5%)
Jorm IQCODE score from Core (Mean (SD))	3.8 (0.8)
Jorm IQCODE score from Core (missing)	3,283

Source: R/MS_MAIN/MS_001_analysis_cohort.R; R_objects/A7_005_hrs16_merged.rds.

Jorm IQCODE was collected in the HRS Core interview, not HCAP, and only from participants who did not complete Core cognitive testing.

Notes: The study sample includes all 3,496 HRS/HCAP 2016 participants, including participants without Core cognitive testing who were classified from informant (Jorm IQCODE) data. Race and ethnicity data were gathered via self-report at the time of first interview and are considered to be markers of exposure to evolving systems of racism, not as a proxy for genetic variation or any other biological variable. Race was self-selected by participants at the time of the first interview from a list of options defined by the 2000 US Census criteria. Other race includes a pooled group of participants who identified as American Indian or Alaska Native, Asian, Native Hawaiian or Pacific Islander, or another self-described race, consolidated due to small sample sizes and risk of identification.

Table 2. HCAP classification margins by method

	Normal		MCI		Dementia	
Method	N	Weighted %	N	Weighted %	N	Weighted %
HRS/HCAP Algorithm	2,313	69.4%	808	21.4%	375	9.2%
Core algorithm	2,385	69.6%	767	21.2%	344	9.1%
PMM	2,557	76.7%	568	13.6%	371	9.7%
Secondary comparators						
Langa-Weir	2,432	74.9%	731	17.3%	333	7.9%
Hudomiet	2,326	71.1%	865	21.0%	286	7.2%

Note. N values are unweighted HRS/HCAP participant counts. Percentages use HCAP16WGTR; PMM counts and percentages use the non-replicated combined cognition/Jorm model sample.

References. Manly, J. J., Jones, R. N., Langa, K. M., Ryan, L. H., Levine, D. A., McCammon, R., Heeringa, S. G., & Weir, D. (2022). Estimating the prevalence of dementia and mild cognitive impairment in the US: The 2016 Health and Retirement Study Harmonized Cognitive Assessment Protocol Project. JAMA Neurology, 79(12), 1242-1249. https://doi.org/10.1001/jamaneurol.2022.3543

Langa, K. M., Weir, D. R., Kabeto, M., & Sonnega, A. (2023). Langa-Weir Classification of Cognitive Function (1995-2020). https://hrsdata.isr.umich.edu/sites/default/files/documentation/data-descriptions/1695907706/Data_Description_Langa_Weir_Classifications2020_V2.pdf

Hudomiet, P., Hurd, M. D., & Rohwedder, S. (2022). Trends in inequalities in the prevalence of dementia in the United States. Proceedings of the National Academy of Sciences, 119(46), e2212205119. https://doi.org/10.1073/pnas.2212205119

Table 3. Agreement among Core classifications and comparison diagnoses

Classification 1	Classification 2	Three-class quadratic weighted kappa	Dementia versus non-dementia kappa
Core Algorithm	HRS/HCAP Algorithm	0.501	0.337
PMM	HRS/HCAP Algorithm	0.509	0.429
Langa-Weir	HRS/HCAP Algorithm	0.510	0.407
Hudomiet	HRS/HCAP Algorithm	0.551	0.457
HCAP Consensus	HRS/HCAP Algorithm	0.728	0.536
Core Algorithm	Langa-Weir	0.563	0.406
PMM	Langa-Weir	0.575	0.480
Hudomiet	Langa-Weir	0.717	0.670
HCAP Consensus	Langa-Weir	0.099	0.736
Core Algorithm	Hudomiet	0.607	0.442
PMM	Hudomiet	0.623	0.529
HCAP Consensus	Hudomiet	0.396	0.806
Core Algorithm	HCAP Consensus	0.083	0.153
PMM	HCAP Consensus	0.486	0.464

Source: A7 rows: R_objects/A7_100_hcap_tables.rds, generated by R/A7_100-comparison_of_diagnoses.R. PMM rows: R/PMM_110_Comparison_of_Consensus_Langa_Weir.R; PMM-Hudomiet was calculated from the same saved probabilities and A7 comparison data. Three-class values are quadratic weighted kappas; binary values are kappas for dementia versus non-dementia. Consensus comparisons use normalized HCAP16WGTR / samplingP weights; other comparisons use HCAP16WGTR.

Table 4. Pairwise three-class agreement among HCAP comparison methods

method	Consensus diagnosis	Algorithmic diagnosis	Langa-Weir	HRS classification model	Hudomiet
Consensus diagnosis	1.000	0.728	0.099	0.083	0.396
Algorithmic diagnosis	0.728	1.000	0.510	0.501	0.551
Langa-Weir	0.099	0.510	1.000	0.563	0.717
HRS classification model	0.083	0.501	0.563	1.000	0.607
Hudomiet	0.396	0.551	0.717	0.607	1.000

Source: Source: R_objects/A7_100_hcap_tables.rds, generated by R/A7_100-comparison_of_diagnoses.R. Entries are quadratic weighted kappas. Consensus comparisons use normalized HCAP16WGTR / samplingP weights; other comparisons use HCAP16WGTR.

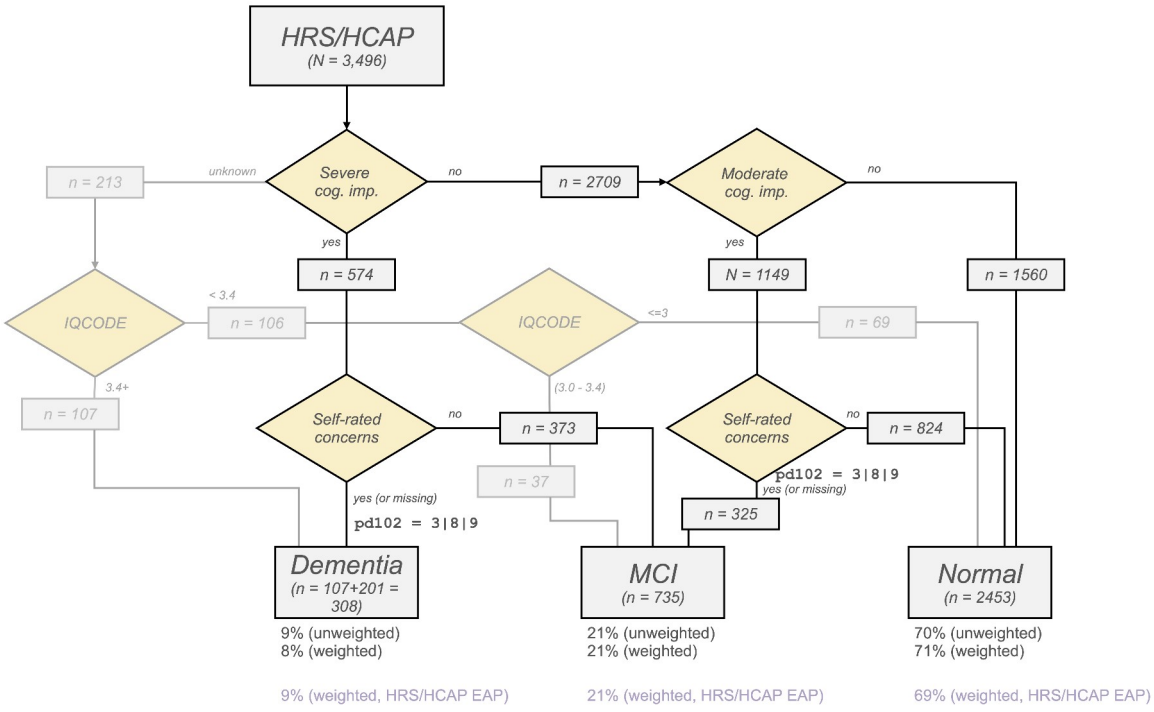


Figure 1: Core algorithm

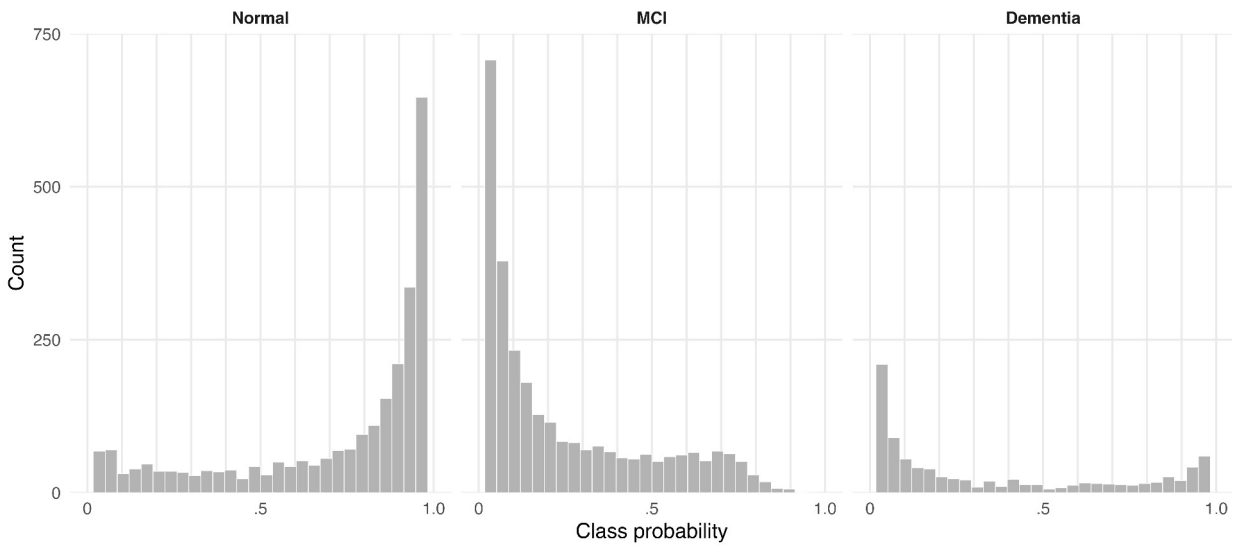


Figure 2: PMM class probabilities in HRS/HCAP

Thanks for reading.

Adaptation of the Manly-Jones algorithm to the HRS core survey data

2026-10-17

Purpose and overview

The analysis of the incidence of MCI and dementia in the HRS-HCAP Wave 1 sample is supported by a cognitive classification at the last HRS core interview for participants who died or dropped out before HRS-HCAP Wave 2 (2022). The Manly et al. (2022) algorithm cannot be applied to core interview data as written, as it classifies participants as having normal cognition, mild cognitive impairment (MCI), or dementia from 5 cognitive performance domain scores estimated from an hour-long neuropsychological battery, together with an informant interview on cognitive and functional change (Langa et al., 2020). The HRS core interview, which has measured cognition in older adults every 2 years since the 1990s with brief tests (Crimmins et al., 2011; Ofstedal et al., 2005), has a brief battery of scored cognitive items (8 used here), 1 self-rated memory item, and an informant questionnaire that is only administered when a proxy answers for the respondent and no cognitive performance data is available. These cognitive measures are too few to score separate domains defined in HCAP, and the lack of informant reports on the daily function of a respondent who answers for themselves generates an important additional difference in available information. To address these differences, we constructed an adaptation we call the “core algorithm”, that follows the logic of the Manly-Jones algorithm with core measures and is calibrated to the Manly-Jones classifications of the same people at HCAP Wave 1 (2016).

The core algorithm has 4 steps (Figure A1): (1) a single cognitive factor score is estimated from the core items; (2) the factor score is normed against the HRS-HCAP robust normative sample and expressed as a demographically adjusted T score; (3) decision rules combine the T score with self-rated memory change, or use the informant questionnaire when no cognitive tests were completed; using (4) T-score cut points set so that the weighted class shares match those of the Manly-Jones algorithm at HCAP Wave 1 (2016). Every parameter (item parameters, norms, and cut points) was estimated once using core data in the HRS-HCAP Wave 1 sample, and then held fixed and used for the classification of participants in the 2016, 2018, 2020, and 2022 core interviews. A change in a participant’s classification across waves will reflect changes in that participant’s responses or age, but the scoring model does not change. The same item thresholds, norms, and cut points convert a 2022 response pattern exactly as they would convert the same pattern observed in 2016.

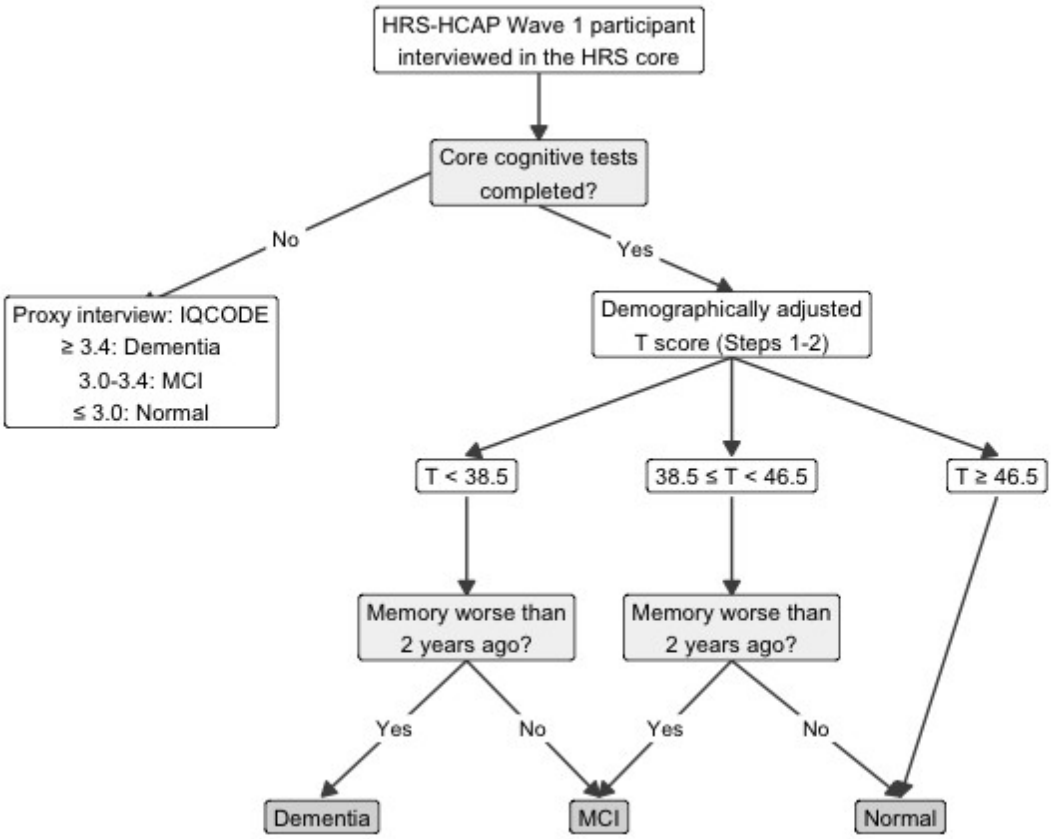


Figure A1. Core algorithm decision structure. Cut points are those selected in Step 4; counts at each node are in Table A3. IQCODE: Informant Questionnaire on Cognitive Decline in the Elderly. Memory worse includes don't know, refused, and not ascertained.

Data

The analytic sample is the 3,496 HRS-HCAP Wave 1 (2016) participants, followed in the HRS core interviews of 2016, 2018, 2020, and 2022. The core algorithm uses 4 kinds of core data: (1) the 8 cognitive items in Table A1; (2) self-rated memory compared with 2 years ago (HRS item D102: better, about the same, or worse); (3) the 16-item informant questionnaire (IQCODE; Jorm, 1994), scored as the mean item rating from 1 (much improved) to 5 (much worse) and collected in proxy interviews; and (4) age, sex, race and ethnicity (non-Hispanic Black, Hispanic), and years of schooling. The calibration target for the core algorithm is the Manly-Jones classification algorithm as obtained in the HRS-HCAP Wave 1 sample. The Langa-Weir classification from the 2016 core interview (Crimmins et al., 2011) is used as a comparator used in the manuscript's sensitivity analysis, and we report its agreement with the Manly-Jones algorithm alongside that of the core algorithm.

Step 1. A single cognitive score from the core battery

The Manly-Jones algorithm counts impaired domains among 5 factor scores (memory, executive function, language, visuospatial ability, and orientation), three of which (memory, executive function, language) are estimated from several HRS-HCAP tests (Jones et al., 2024) and four of which are demographically adjusted (orientation is not). The core battery has at most 3 items for any domain, no item for visuospatial ability (the battery was first built for telephone administration; Ofstedal et al., 2005), and a mix of item formats (counts with ceilings, a pass/fail item, and continuous scores). In the core data, the 5 domain scores cannot be estimated with useful precision.

We addressed this with a single-factor confirmatory factor analysis (CFA) model that treats every item on its own scale. Ordinal and binary items entered through probit thresholds, and the 3 continuous scores (animal naming, delayed word recall, and number series) were rescaled to the 0-1 range (continuous) of 2016 responses. The model was fit to HRS 2016 respondents aged 65 and older (N = 9,218) with the complex sample design (64 strata, 121 clusters) and the HRS-HCAP Wave 1 weights. Respondents outside HRS-HCAP received a weight of 0, so the estimates describe the population that HRS-HCAP represents (3,283 HRS-HCAP participants with at least 1 HRS core item response). Model fit was evaluated in the pairwise complete sample with the weighted least squares mean and variance adjusted (WLSMV) estimator using Mplus software (version 9, Muthén & Muthén, Los Angeles, CA). Item parameters were then estimated by maximum likelihood with robust standard errors (MLR) and a probit link, which uses every available item response. In subsequent factor score estimation runs, all parameters were fixed at the HCAP Wave 1 estimates. Expected a posteriori (EAP) factor scores (Bock & Mislevy, 1982) were estimated from the fixed model for every core person-wave from 2016 to 2022 with at least 1 item response (63,992 person-waves, all ages). As with our analysis of the HCAP battery, immediate word recall was not included in the model because it adds little information beyond delayed recall in the HCAP Wave 1 sample (Jones et al., 2024).

The single-factor model fit well (RMSEA = 0.035, 90% CI 0.031 to 0.039; CFI = 0.966; TLI = 0.952; SRMR = 0.047)[1], by the conventional guides of Hu and Bentler (1999). Standardized

[1] RMSEA, root mean square error of approximation, estimates model misfit per degree of freedom in the population and does not compare the model with a baseline model. Lower values indicate better fit; values below 0.05 indicate close fit (Hu & Bentler, 1999; Steiger, 1990). CFI, comparative fit index, and TLI, Tucker-Lewis index, compare the fitted model with a baseline model in which the items are uncorrelated. Higher values indicate better fit; values of 0.95 or above indicate satisfactory fit (Bentler, 1990; Hu & Bentler, 1999; Tucker & Lewis, 1973). The TLI adds a penalty for model complexity. SRMR, standardized root mean square residual, is the average difference between the observed and model-implied correlations. Lower values indicate better fit; values of 0.08 or below indicate satisfactory fit (Asparouhov & Muthén, 2018; Hu & Bentler, 1999).

loadings in the MLR model ranged from 0.48 to 0.71 (Table A1), and that range indicates that no single item dominates the factor. Table A1 provides loadings from the MLR and WLSMV models. Loadings from the two estimators were similar for most items. The lower WLSMV loading for number series, the item with the most missing data (12% of HRS-HCAP participants with any core item response, against at most 1% for any other item), likely reflects how each estimator handles missing data: WLSMV estimates item correlations from pairwise-present data and assumes data are missing completely at random, whereas MLR uses all available responses and assumes data are missing at random. The higher WLSMV loading for object naming cannot be explained by missing data, which was nearly complete for that item; we did not examine the difference further, but it may be related to the item's skewed response distribution (93% of HRS-HCAP participants named both objects correctly, weighted).

Table A1. HRS core cognitive items and standardized factor loadings

Item	Scoring	Loading (MLR, used for scoring)	Loading (WLSMV)
Orientation to date	Month, day, year, day of week; 0-4 correct (ordinal)	0.48	0.52
Animal naming	Animals named minus errors; rescaled to 0-1 (continuous)	0.60	0.63
Object naming (scissors, cactus)	0-2 correct (ordinal)	0.68	0.84
Naming the President and Vice President	0-2 correct (ordinal)	0.60	0.64
Delayed word recall	0-10 words; rescaled to 0-1 (continuous)	0.63	0.64
Number series	Score rescaled to 0-1 (continuous)	0.71	0.62
Serial 7 subtraction	0-5 correct (ordinal)	0.66	0.60
Backward counting from 20	Correct on first or second try (binary)	0.52	0.56

Single-factor model fit to HRS 2016 respondents aged 65 and older (N = 9,218) with HRS-HCAP weights (HCAP16WGTR; weight 0 outside HRS-HCAP), 64 strata and 121 clusters. Loadings are standardized (STDYX). Continuous items were rescaled to 0-1 using the 2016 range. WLSMV fit: chi-square = 248.5 (df = 20), RMSEA = 0.035, CFI = 0.966, TLI = 0.952, SRMR = 0.047. Source: R/A7_050-mplus_model.R.

Step 2. Demographic norms

The Manly-Jones algorithm defines impairment as performance more than 1.5 standard deviations below the level expected for a person's age, sex, race, ethnicity, and education in a robust normative sample (Manly et al., 2022). Although we use different thresholds to define impairment (discussed below), we norm the single general cognitive domain score the same way we normed domain scores in Manly et al. (2022).

The robust normative sample (n = 1,787) is used to define “expected” cognitive performance levels. Factor scores in the normative sample were converted to normal scores with the Blom (1958) rank transformation. Because this transformation is sample dependent, we derive a linking equation between Blom transformed scores and estimated factor scores using linear regression with a restricted cubic spline for the factor score (knots at the 5th, 35th, 65th, and 95th percentiles; Harrell, 2015). This regression converts any factor score, including scores from people outside the normative sample and from later waves, to the normalized metric.

Then, normalized score was regressed, with survey weights, on age (restricted cubic spline with knots at 70, 78, 86, 94 years, centered at 70), sex, non-Hispanic Black race, Hispanic ethnicity, and years of schooling (centered at 12), together with their 2-way interactions (24 terms; R^2 = 0.37). This demographic equation is used to define the expected score.

Finally, the T score was computed as 50 + 10 × (observed − expected) / 0.775, where the denominator is the standard error of the estimate from the demographic regression in the normative sample (the SD of the normalized score, 0.973, multiplied by $\sqrt{(1 - R^2)}$).

A T score of 40 therefore means performance 1 residual standard deviation (0.775 normalized-score units) below the level expected for a person of the same age, sex, race, ethnicity, and education; the same residual standard deviation is assumed for every demographic group. In HRS-HCAP Wave 1, the unweighted mean (SD) T score was 50 (10) in the normative sample and 43 (12) among participants excluded from it. Age is updated at each wave, so the expected score generally falls as a participant ages. A participant whose factor score declines at the rate implied by the 2016 cross-sectional age gradient keeps a stable T score, and an incident impairment classification requires decline faster than that gradient.

Step 3. Classification rules

In the Manly-Jones algorithm, dementia requires at least 2 impaired domains and an informant report of functional impairment, and MCI requires at least 1 impaired domain with informant-reported concern or self-reported memory worsening (Manly et al., 2022). The core interview has no domains, and it has no informant report for respondents who answer for themselves.

We replaced each element with its closest core counterpart (Table A2). Two T-score cut points stand in for the domain count: T below the severe cut point corresponds to impairment in 2 or more domains, and T between the severe and moderate cut points corresponds to impairment in 1 domain. Self-rated memory worsening, the item the Manly-Jones algorithm already uses for MCI, takes the place of informant report at both levels. Participants without core cognitive tests

(mostly proxy interviews) were classified from the IQCODE with the Manly-Jones informant thresholds (3.4 or higher for dementia; above 3.0 for MCI). Missing self-rated memory was singly imputed by chained equations (van Buuren & Groothuis-Oudshoorn, 2011); no Wave 1 participant with a T score was missing it, and “don’t know,” refused, and not ascertained responses (4 participants) were counted as worse, which assigns these participants to the more impaired class.

The memory rule decides the class of most participants with low T scores. Of the 724 Wave 1 participants with T below 38.5, 487 (67.3%) reported that their memory was the same as or better than 2 years earlier and were classified as having MCI rather than dementia (Table A3). This pattern may reflect reduced awareness of memory loss among people with dementia, a possibility these data cannot test; either way, the rule likely moves some people who meet the Manly-Jones criteria for dementia into the MCI class.

Table A2. Core algorithm decision rules

Route	Cognitive criterion	Second criterion	Classification
Cognitive tests	T < 38.5	Memory worse than 2 years ago*	Dementia
Cognitive tests	T < 38.5	Memory better or the same	MCI
Cognitive tests	38.5 ≤ T < 46.5	Memory worse than 2 years ago*	MCI
Cognitive tests	38.5 ≤ T < 46.5	Memory better or the same	Normal
Cognitive tests	T ≥ 46.5	Any	Normal
Proxy interview	No T score	IQCODE ≥ 3.4	Dementia
Proxy interview	No T score	3.0 < IQCODE < 3.4	MCI
Proxy interview	No T score	IQCODE ≤ 3.0	Normal

T: demographically adjusted T score (mean 50, SD 10 in the HRS-HCAP robust normative sample). *Self-rated memory compared with 2 years ago (HRS item D102); don't know, refused, and not ascertained count as worse; missing values were singly imputed. IQCODE: 16-item Informant Questionnaire on Cognitive Decline in the Elderly, mean item score (1-5), asked in proxy interviews. Source: R/A7_060-algorithm_functions.R; R/A7_075-implement_algorithm.R.

Table A3. Participants at each node of the core algorithm, HRS-HCAP Wave 1 (2016)

Node	n
HRS-HCAP Wave 1 participants	3,496
No core cognitive tests (IQCODE route)	213
IQCODE: Dementia	107
IQCODE: MCI	37
IQCODE: Normal	69
T < 38.5	724

Table A2. Core algorithm decision rules

Route	Cognitive criterion	Second criterion	Classification
Memory worse: Dementia			237
Memory not worse: MCI			487
38.5 ≤ T < 46.5			835
Memory worse: MCI			243
Memory not worse: Normal			592
T ≥ 46.5: Normal			1,724
Total classified Normal			2,385
Total classified MCI			767
Total classified Dementia			344

Unweighted counts. Source: R/A7_075-implement_algorithm.R.

Step 4. Cut points by prevalence matching

The fourth problem is where to place the cut points. No point on a single T score corresponds by construction to “2 or more domains more than 1.5 standard deviations below expectation,” because the single score pools information that the Manly-Jones algorithm keeps separate. The cut points must be set empirically.

We set them by prevalence matching: the cut points were chosen so that the weighted Normal, MCI, and dementia shares of the core algorithm match those of the Manly-Jones algorithm in HRS-HCAP Wave 1 (N = 3,496). We evaluated every pair of cut points on a grid (severe cut point 25 to 50, moderate cut point 30 to 60, in steps of 0.5), applying the memory and IQCODE rules unchanged, and selected the pair that minimized the sum of squared differences between the 3 core and Manly-Jones shares, with ties broken by the higher weighted kappa. The dementia share depends only on the severe cut point, and the MCI share depends on both, so the search amounts to reading 2 curves (Figure A2). We matched shares, rather than maximizing agreement, because incidence and prevalence estimates depend on class shares; matching keeps population estimates from core data on the same footing as estimates from HRS-HCAP data. Matching shares does not guarantee that the same people are classified alike, and we report agreement separately below.

The selected cut points were T < 38.5 (severe) and T < 46.5 (moderate). With these cut points the weighted shares were 69.6% normal, 21.2% MCI, and 9.1% dementia, against 69.4%,

21.4%, and 9.2% for the Manly-Jones algorithm (Table A4). The cut points used before this calibration (T < 37 and T < 48) gave 20.7% MCI and 8.3% dementia. The pair with the highest 3-class kappa (0.52, at T < 36 and T < 42.5) gave 14.8% MCI and 7.8% dementia. The cut points were chosen and evaluated in the same sample, so the agreement statistics in Tables A4 and A5 are likely optimistic.

Table A4. T-score cut points: selected, previous, and kappa-maximizing

Set	Severe cut	Moderate cut	Normal, %	MCI, %	Dementia, %	Kappa, 3-class	Kappa, dementia
Selected (prevalence matching)	38.5	46.5	69.6	21.2	9.1	0.50	0.34
Previous	37	48	71.0	20.7	8.3	0.50	0.35
Highest 3-class kappa	36	42.5	77.4	14.8	7.8	0.52	0.35
Target: Manly-Jones algorithm			69.4	21.4	9.2		

HRS-HCAP Wave 1, N = 3,496. Percentages and kappas are weighted (HCAP16WGTR). Kappas compare the core algorithm with the Manly-Jones algorithm (3-class: quadratic weights; dementia: dementia versus not). Grid step 0.5 T-score points. Cut points were chosen and evaluated in the same sample. Source: R/A7_070-threshold_calibration.R.

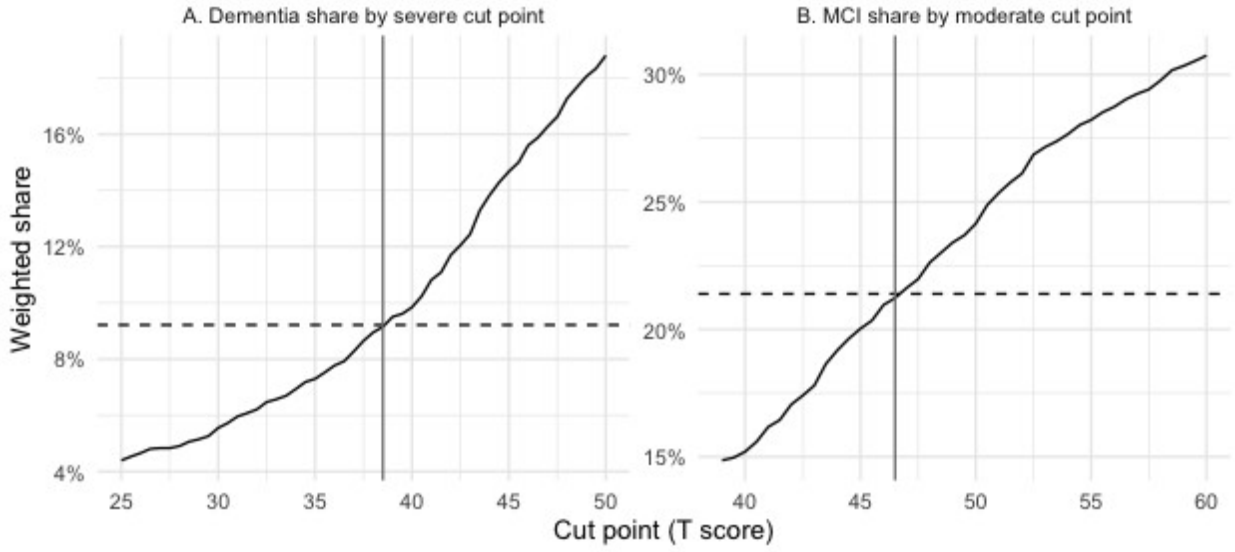


Figure A2. Weighted class shares of the core algorithm across cut points, HRS-HCAP Wave 1. Panel A: dementia share by severe cut point. Panel B: MCI share by moderate cut point, at the selected severe cut point. Dashed lines: Manly-Jones shares. Vertical lines: selected cut points.

Agreement with the Manly-Jones algorithm in HRS-HCAP Wave 1

For the 3 classes, agreement of the core algorithm with the Manly-Jones algorithm was about the same as the agreement of Langa-Weir; for dementia versus not, it was lower than that of Langa-Weir (Table A5). The quadratic weighted kappa for the 3 classes was 0.50 for the core algorithm and 0.51 for Langa-Weir (Cohen, 1968); for dementia versus not, kappa was 0.34 and 0.41. Of the weighted population the Manly-Jones algorithm classified as having dementia, the

core algorithm classified 40% as having dementia and Langa-Weir classified 42%. Of those the core algorithm classified as having dementia, 40% had dementia by the Manly-Jones algorithm (Langa-Weir: 50%). Agreement between the core algorithm and Langa-Weir was 0.56 (3 classes) and 0.41 (dementia). Each kappa compares 2 classifications that both contain error, so these values describe agreement and not accuracy; agreement with clinical consensus diagnoses in HRS-HCAP is reported elsewhere (Farron et al., 2026).

Table A5. Weighted cross-classification of the core algorithm and Langa-Weir with the Manly-Jones algorithm, HRS-HCAP Wave 1

Comparator	Manly-Jones class	Comparator class: Normal	Comparator class: MCI	Comparator class: Dementia	Total
Core algorithm	Normal	28,842 (83%)	4,941 (14%)	1,005 (3%)	34,788
Core algorithm	MCI	5,262 (49%)	3,712 (35%)	1,751 (16%)	10,725
Core algorithm	Dementia	798 (17%)	1,991 (43%)	1,831 (40%)	4,620
Langa-Weir	Normal	30,246 (87%)	4,091 (12%)	450 (1%)	34,788
Langa-Weir	MCI	6,070 (57%)	3,120 (29%)	1,535 (14%)	10,725
Langa-Weir	Dementia	1,220 (26%)	1,439 (31%)	1,961 (42%)	4,620

Rows: Manly-Jones class (HRS-HCAP). Columns: comparator class (core algorithm or Langa-Weir) from 2016 HRS core data. Cell entries are survey-weighted population counts in thousands. Entries for fewer than 1,000 persons show two significant digits. Row percentages in parentheses. N = 3,496 participants with both classifications (HCAP16WGTR). Quadratic weighted kappa (3-class): core 0.50, Langa-Weir 0.51. Kappa (dementia versus not): core 0.34, Langa-Weir 0.41. Source: R/A7_100-comparison_of_diagnoses.R.

Application to the 2018, 2020, and 2022 core interviews

The fixed model, norms, and cut points were applied to every core interview of HRS-HCAP Wave 1 participants from 2016 to 2022 (Table A6). The number of classified person-waves fell from 3,496 in 2016 to 2,049 in 2022 as participants died or left the study, and the share classified from the IQCODE alone was 6.1% in 2016 and 6.9% in 2022.

One wave departs from the others. Number series was not administered in the 2018 core interview (no 2018 respondent has a score), so 2018 EAP scores rest on 7 items. With less information, an EAP score is pulled further toward the population mean (Bock & Mislevy, 1982), which raises the scores of low performers and may classify fewer participants as impaired in 2018 than the same performance would produce in other waves. We did not re-norm the 7-item score. In the incidence analysis, 2018 is the last classified interview for participants who died or dropped out between the 2018 and 2020 interviews, and their final classification carries this bias.

Table A6. Core algorithm classifications of HRS-HCAP Wave 1 participants interviewed in the HRS core, 2016-2022

Core wave	Person-waves	Cognitive tests	IQCODE only	Not classified	Normal	MCI	Dementia
2016	3,496	3,283	213	0	2,385 (68.2%)	767 (21.9%)	344 (9.8%)
2018	2,989	2,836	152	1	2,135 (71.5%)	606 (20.3%)	247 (8.3%)
2020	2,558	2,380	178	0	1,891 (73.9%)	466 (18.2%)	201 (7.9%)
2022	2,050	1,907	142	1	1,468 (71.6%)	380 (18.5%)	201 (9.8%)

Unweighted counts of person-waves; percentages among classified person-waves. Not classified: no cognitive tests and no IQCODE. Number series was not administered in the 2018 core interview, so 2018 T scores rest on 7 items. Source: R/A7_075-implement_algorithm.R.

Limitations

Limitations of our approach are worth mentioning. First, the cut points were selected and evaluated in the same HRS-HCAP Wave 1 sample, so the agreement in Tables A4 and A5 is in-sample. The comparison of core-based and HRS-HCAP-based classifications among Wave 2 re-interviewees (eTables 7 and 8) is the out-of-sample check available in these data. Second, self-rated memory replaces informant report, and most participants with low T scores did not report memory worsening; the core algorithm likely under-classifies dementia relative to the Manly-Jones algorithm, which is consistent with the lower core-based dementia incidence among re-interviewees in eTable 8. Third, the 2018 scores rest on 7 items (above). Fourth, the norms do not account for interview mode, which varies across respondents and waves; a mode difference in test scores would shift T scores in the waves where the mode mix differs from 2016. Fifth, core respondents take the same tests every 2 years, but by 2016 respondents aged 65 and older had already completed several prior core assessments, and the normative sample shares that history, so the added practice effect from 2016 to 2022 is likely small. Sixth, the IQCODE in the core interview asks about change over 2 years, whereas the HRS-HCAP informant interview asks about change over 10 years, so the Manly-Jones IQCODE thresholds may not carry the same meaning in core proxy interviews. Seventh, the kappas are point estimates; we have not yet estimated their sampling error under the complex design.

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Appendix 2. PMM and logistic regression

2026-10-17

PMM and logistic regression can both classify people into Normal, MCI, or Dementia. The difference is what they model.

Logistic regression

Logistic regression starts with a known outcome, Y, and predicts it from observed predictors, X:

$$P(Y = \text{Dementia} \mid X)$$

For example, it could predict dementia from word recall, serial sevens, ADLs, and Jorm score. It estimates how each predictor changes the log-odds of each outcome category. In a multinomial logistic model, the usual form is:

$$\log \frac{P(Y = k \mid X)}{P(Y = \text{Normal} \mid X)} = \beta_{0k} + \beta_{1k} X_1 + \cdots + \beta_{pk} X_p$$

The main modeling decision is the regression equation: which predictors, interactions, nonlinear terms, and coefficients to include.

PMM in this project

The PMM takes a generative approach. It asks: *If a person belongs to the Normal, MCI, or Dementia class, what pattern of cognitive-test, functional, self-report, and informant responses would we expect?* PMM models:

$$P(X \mid C = k)$$

for each cognitive class $C = k$, along with the class prevalence $P(C = k)$. It then uses Bayes' rule to obtain the classification probability:

$$P(C = k \mid X) = \frac{P(X \mid C = k) P(C = k)}{\sum_j P(X \mid C = j) P(C = j)}$$

So, instead of directly predicting diagnosis from scores, it describes the characteristic response distribution within each diagnosis class and asks which class best accounts for a person's observed pattern.

For the final PMM, the cognition model uses the individual test scores, functional items, and self-report item directly. It estimates class probabilities from that whole response pattern. The separate Jorm PMM does the analogous job for people with informant data rather than cognitive testing.

Practical distinctions

- Logistic regression: “Given these scores, what is the probability of Dementia?”
- PMM: “Which diagnosis class most plausibly generated this pattern of scores and reports?”
- Logistic regression is discriminative: it models $P(Y \mid X)$ directly.
- PMM is generative: it models $P(X \mid C)$, then derives $P(C \mid X)$.
- Logistic regression commonly treats predictors as explanatory variables with regression coefficients.
- PMM treats the observed responses as indicators of an underlying class-specific response profile.
- Logistic regression needs explicit interactions or nonlinear terms when the meaning of one score depends on another.
- A PMM can represent distinct multivariate profiles through class-specific item distributions, although it still relies on assumptions about within-class dependence.

In this project, the important wrinkle is that the PMM is not purely exploratory latent class analysis. It is calibrated using the HRS/HCAP Algorithm classification as the known class in the HCAP sample, then scores the Core measures of the same HCAP participants with the calibrated parameters (including the class proportions) held fixed. That makes it a supervised, model-based classifier: conceptually closer to generative discriminant analysis than to an unsupervised profiling exercise.

A concise manuscript description would be:

> *The PMM is a supervised, generative classifier that estimates the probability of each HRS/HCAP Algorithm class from the joint pattern of observed cognitive, functional, self-report, or informant responses.*

Appendix 3: Comparisons with other study classifications

2026-10-17